DO NOW — RECALL

Lesson 086 · Debugging Unfamiliar Algorithms

Name: _____ Class: _____ Date: _____

1. What is a loop invariant?

2. What is a boundary test?

3. What return value commonly indicates a failed indexed search?

4. In array [2,4,6], set a[1]=a[0]+a[2]. Give the final array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Property maintained after each iteration.

2. Model answer

A test at or next to a threshold.

3. Model answer

−1.

4. Model answer

[2,8,6].

By the end, you can…

• Trace before changing code: list variables, loop indexes, condition and expected return.

• An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

• Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Vocabulary — know these terms

logic error — Program runs but produces wrong answer.

boundary — Edge of valid range.

trace — Record changing states.

debugging — Finding and correcting faults.

NEW LEARNING

Read, annotate and explain

Observe

Trace before changing code: list variables, loop indexes, condition and expected return.

Explain

An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

Evaluate

Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Repair for-loop condition i<=a.Length in an array scan.

Model response

For length4, valid indexes0–3. Checking a[4] would throw.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace CountEven([2,5,8,9]) with count after each item. [6]

Question 2

Explain why i<=a.Length is unsafe in a loop indexing a[i], and repair. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Develop four distinct tests for a function counting even values, with expected answers and reasons. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A loop using i<=Length can safely read a[i] for every i.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.